# ARGO-GIM

INSIDE INSIGHT · GASTROINTESTINAL INTELLIGENCE

CONCEPT & R&D · TALENTIS 2026 BEST RUNNER-UP

## Your gut, seen from the inside. At home, with no invasive preparation.

An ingestible capsule concept combining high-resolution optical cameras, biochemical biosensor arrays, and on-board Edge-AI for longitudinal gut health monitoring and colorectal cancer early prevention.

### CLINICAL & EPIDEMIOLOGICAL PROBLEM

#### Current screening fails to reach those who need it most

Colorectal cancer is the second leading cause of cancer death in Europe. When diagnosed at a localized stage, 5-year survival reaches 91%, but plummets to 13% once metastases occur. In Italy, screening adherence averages only 35.8% (dropping to 21.1% in the South and Islands) due to fear of procedural invasiveness and demanding bowel preparation regimes.

**35.8%**
Italian screening uptake (ONS 2024)

**21.1%**
Uptake in Southern Italy & Islands

**91% / 13%**
5-year survival: localized vs metastatic

### THE ARGO-GIM SOLUTION

#### Integrated 4-component system architecture

ARGO-GIM is designed to transform an invasive hospital procedure into a non-invasive, home-based examination:

- **Miniaturized capsule (28 × 12 mm):** dual-end cameras, biosensor array, ultra-low-power radio, and solid-state biocompatible battery.
- **Lightweight sensor vest:** worn comfortably under daily clothes to track telemetry packets and anatomically localize the capsule.
- **Compact receiver unit:** records physiological signals during normal daily activities.
- **Companion App & AI dashboard:** computes a gut health index and compiles a structured clinical report.

### TECHNOLOGICAL INNOVATION · DIFFERENTIATION FROM CONVENTIONAL CAPSULES (E.G. PILLCAM™)

| FEATURE | CONVENTIONAL ENDOSCOPY CAPSULES (E.G. PILLCAM™) | ARGO-GIM INNOVATION & ADVANTAGES |
|---|---|---|
| **Bowel Preparation** | `Mandatory & Demanding` Intensive laxatives and fasting required to clear optics. | `Non-Invasive` **Biosensors operate even within physiological contents, removing adoption barriers.** |
| **Data Modality** | `Optical video only` Surface photography; blind to biochemical shifts or sub-visual inflammation. | `Multimodal` **High-res optics + biochemical biosensors (pH, oxidative stress ROS, haemoglobin, temp).** |
| **Frame Processing** | `Up to 50,000 raw frames` Hours of manual physician frame-by-frame review. | `On-board Edge-AI` **Local filtering of repetitive frames: transmits only suspicious anomalies, saving time and power.** |
| **Setting of Use** | Hospital specialist outpatient procedure. | **Home-based longitudinal monitoring (Lite) and advanced hospital diagnostics (Dx).** |

### ARGO-GIM LITE · HOME PREVENTION

Tailored for proactive individuals tracking gut health over time. On-board AI identifies baseline shifts, alerting users to seek timely medical consultation before macroscopic lesions develop.

### ARGO-GIM DX · CLINICAL SPECIALIST DEVICE

Equipped with Narrow-Band Imaging (NBI) and structured-light 3D depth mapping. AI-assisted lesion classification (Paris, Kudo, NICE) and native FHIR hospital EHR interoperability.

ARGO-GIM is an early-stage technological and scientific concept, not an approved medical device. Specifications represent design targets to be clinically validated.
Sources: [1] Italian National Screening Observatory (ONS 2024) · [2] American Cancer Society, SEER data · © 2026 ARGO-GIM · info@argogim.com

# ARGO-GIM

EXECUTIVE SUMMARY · 2/2 · DEVELOPMENT ROADMAP & REQUIREMENTS



## TRACK RECORD · RECOGNITIONS & ACCELERATION PROGRAMS

**01 · ACCELERATION & LAB**

**CAL.HUB.RIA - LAB**

Talent, ideas, value for the future: biomedical incubation and scientific validation program.

**02 · INNOVATION CONTEST**

**Talentis · Maratea**

Competitive selection and pitch presentation at the national innovation competition.

**03 · NATIONAL FINALS**

**Talentis · CAPRI 2026**

Awarded as **Best Runner-up**, securing qualification for the national finals in Capri.

## CONCRETE TARGETS · BENCHTOP PROTOTYPE (12 MONTHS)

Our 12-month operational plan targets achieving **TRL 4 (functional benchtop prototype validated in lab)**:

- A **Circuit Integration:** rigid-flex PCB layout integrating ultra-low-power Cortex-M MCU and RF transceiver.
- B **Biosensor Interface:** functional micro-electrodes for pH, haemoglobin, and oxidative stress (ROS) detection.
- C **In-vitro Gut Simulator:** fluidic testing in anatomical phantoms to evaluate buoyancy, optics, and RF transmission through tissue.
- D **Intellectual Property:** patent filing covering the integrated multimodal sensing architecture and Edge-AI filtering pipeline.

## WHAT WE SEEK · TECHNICAL PARTNERSHIPS

To advance the prototype development, we actively seek collaborations with key technical profiles and research partners:

- 01 **TECHNICAL CO-FOUNDER:** Electronic & Biomedical Engineering / Embedded Firmware to lead hardware integration and sensor design.
- 02 **Gastroenterology Units:** clinical partners to co-develop validation protocols and participate in pilot studies.
- 03 **MedTech Regulatory Mentors:** guidance for EU MDR 2017/745 pathway and ISO 10993 biocompatibility standards.
- 04 **Microelectronic Laboratories:** research partners offering testing benches and anechoic chambers.

## REGULATORY PATHWAY & DEVELOPMENT PHASES (EU MDR 2017/745)

**PHASE 1 · TODAY**
**Concept & Design**
3D CAD, simulations, and sensor selection.

**PHASE 2**
**TRL 4 Prototype**
Hardware integration, in-vitro testing, and patent.

**PHASE 3**
**Preclinical**
ISO 10993 biocompatibility & electrical safety.

**PHASE 4**
**Clinical Trials**
Feasibility and comparative clinical validation.

**PHASE 5**
**CE Marking**
Medical device clearance for Lite & Dx.

## THE ARGO-GIM DEVELOPMENT TEAM

**NV** **Niccolò Vono** · CO-FOUNDER
PhD Student in Molecular Medicine Biotechnology · colorectal cancer multi-omics

**CV** **Cristina Vocca** · CO-FOUNDER
PhD Student in Public Health · resident in Clinical Pharmacology and Toxicology

**GF** **Giuseppina Fruci** · CO-FOUNDER
Master's Student in Pharmacy

## DIRECT CONTACT CHANNEL

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